

Sahil Dangol

Full-Stack Gen AI Engineer | FinTech & Quantitative Research Enthusiast
Lalitpur, Nepal | sahil.info.np | sahil-dangol

PROFESSIONAL SUMMARY

FinTech-focused FullStack AI Engineer specializing in Generative AI, Software Engineering, and Microservices architecture. Expert in building high-performance backend systems for financial reporting, real-time distributed ecosystems, and predictive equity models. Proven record in leading engineering teams to architect and deliver enterprise-grade accounting software.

TECHNICAL SKILLS

Data & AI Engineering: Agentic Workflows, RAG, CAG, LoRA Fine-tuning, PyTorch, CatBoost, Temporal Fusion Transformers, Pandas, NumPy

Backend & Architecture: Python (FastAPI, Django), Node.js (Nest.js), Microservices, SQL (PostgreSQL), NOSQL, Redis

Tools & DevOps: Bruno, Postman, Docker, Azure Serverless, CI/CD, WebSockets, Git, CDN, BLE GATT

Frontend & Mobile: Vue.js, Next.js, React, Tailwind CSS, Flutter, Dart

FINTECH PROJECTS

Quantitative NEPSE Stock Predictor | *CatBoost, Financial Data Engineering, Python*

- Developed a predictive quantitative model using CatBoost to forecast short-term equity returns in the Nepalese market.
- Engineered a dynamic market regime filter using 20-day SMA to optimize entry signals and mitigate risk during bearish cycles.
- Implemented 9 high-signal temporal features and used log-returns to ensure statistical stationarity in financial time-series.

AutoTFT Financial Forecasting Research | *PyTorch, PyTorch Lightning, Scikit-Learn*

- Leading research on Temporal Fusion Transformers (TFT) to predict 7-day price movements for the banking sector.
- Curating a 1-year longitudinal dataset integrating OHLCV data with fundamental metrics (ROE, PBV, NPL) for Commercial Banks.

Agentic AI Therapist | *Quantized Open-Source LLM, Gemini API, Hybrid RAG, FastAPI, Docker*

- Architected a zero-knowledge AI system using local inference and Gemini APIs for advanced cognitive reasoning.
- Designed privacy-centric pipelines for conversational data, ensuring data sovereignty through custom anonymization layers.

EXPERIENCE

Asteroid Studios | *Software Engineer*

May 2024 – Present

- Developed the end-to-end development of enterprise accounting features, designing SQL structures for complex financial reporting.
- Architected a hardware-integrated ecosystem and scalable microservices to support high-concurrency data transactions.
- Bridged technical execution with business strategy for a team of 5+, focusing on low-latency communication and system integrity.

KathFOSS (Open Source Community) | *President*

2025 – Present

- Managing Nepal's prominent open-source hub, facilitating national collaborations between industry and academia.
- Supervised the delivery of scalable open source web platforms and projects using Agile (Scrum/Kanban) methodologies.

EDUCATION & CERTIFICATIONS

Tribhuvan University, Nepal | *Bachelor of Computer Engineering*

2023 – 2027 (Apr)

- Distinction: Top 5% of Cohort; Applying rigorous engineering principles and meticulous attention to detail to architect highly scalable and fault-tolerant systems.

Rice University, USA | *Engineering Project Management Specialization*

2025

- Advanced training in SDLC, Risk Mitigation, and Stakeholder Management for engineering projects.

AWARDS & RECOGNITIONS

- NASA Space Apps Hackathon (2023):** Global Nominee. Architected an immersive terrain engine with custom landing physics.
- Nationwide Hackathon Winner (2024):** Developed Krishi Sathi, an AI assistant with localized financial planning modules.